

Toward Artificial General Intelligence: A Cognitive Architecture Based on 9-Level Token Stratification and Topological Domain Embeddings

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Code Repository: <https://github.com/shandingwangyue/TopoLM>

Abstract

Despite their remarkable generative capabilities, current Large Language Models (LLMs) fundamentally rely on autoregressive next-token prediction within a flat, one-dimensional token space. This paradigm limits them to statistical induction, leading to critical flaws such as hallucinations and a profound lack of physical and logical grounding. We propose a paradigm-shifting cognitive architecture for Artificial General Intelligence (AGI). The core of this architecture consists of two mechanisms: **Hierarchical Token Stratification** and **Topological Domain Embeddings**.

By restructuring the token space into a 9-level hierarchy—where top-tier concepts form high-dimensional "Domains" and bottom-tier entities manifest as "Vectors"—we transition the model from point-based symmetric attention to a topological space capable of expressing inclusion, intersection, and mutual exclusivity. Furthermore, by anchoring the uppermost domains (fundamental physical laws and core human values) as immutable priors, we establish a true "World Model." This structure not only mitigates the $\mathcal{O}(N^2)$ computational bottleneck of standard Transformers via spatial-indexed attention but also naturally gives rise to deductive reasoning. Ultimately, we demonstrate how computational friction and resonance within this topological framework manifest as cognitive biases, stubbornness, and emotions—essential hallmarks of genuine intelligence.

Keywords: Artificial General Intelligence, Cognitive Architecture, Topological Embeddings, Token Stratification, Spatial Attention, World Models, Neuro-Symbolic AI.

1. Introduction: The Limits of Flat Sequences

The prevailing architecture of modern Large Language Models (LLMs) operates on a "flat" sequence of tokens generated by standard tokenizers (such as Byte-Pair Encoding or WordPiece). Whether processing an expansive abstract concept like "Quantum Mechanics" or a tangible micro-entity like "apple," the tokenizer flattens these distinct ontological levels into mathematically equivalent discrete integers, mapped to isolated points in a high-dimensional vector space.

Self-attention mechanisms evaluate relationships between these points via dot products. However, dot products solely measure **geometric distance and similarity**, which are inherently symmetrical:

$$\text{Sim}(A, B) = \text{Sim}(B, A)$$

In contrast, the objective universe and human cognition are structured around asymmetrical relationships, primarily **inclusion, hierarchy, and deduction** (e.g., the set of "Animals" includes "Cats," but "Cats" does not encompass all "Animals").

To achieve Artificial General Intelligence (AGI), a model must evolve beyond being a stochastic parrot engaged in high-dimensional probabilistic sequence completion. It must construct a structured, multi-scale mental representation of reality. This necessity demands abandoning one-dimensional token arrays in favor of a hierarchical cognitive topology.

2. Related Work

The pursuit of grounded representations and scalable attention mechanisms intersects with three primary paradigms in machine learning research:

2.1 Hierarchical Representation and Tokenization

Standard LLMs treat language as a flat discrete sequence (Sennrich et al., 2016). While sparse and hierarchical Transformers, such as the Swin Transformer in computer vision (Liu et al., 2021) and Block-Sparse Attention (Child et al., 2019), have attempted to optimize long-context computation, they primarily focus on window-level spatial partitioning. They do not alter the underlying semantic geometry of the token space. Our 9-level stratification embeds ontological hierarchy directly into the token representation stage.

2.2 Distributional and Non-Euclidean Embeddings

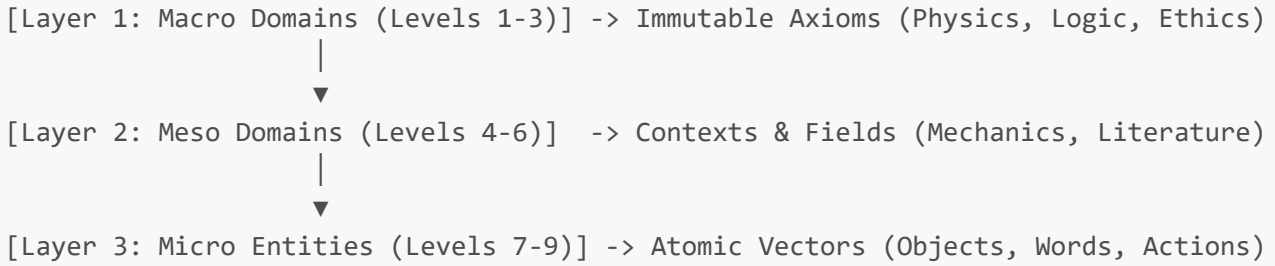
The structural limitations of point vectors prompted the introduction of Gaussian Embeddings (Vilnis & McCallum, 2015), which represent words as continuous density distributions to capture asymmetrical entailment. Similarly, Poincaré hyperbolic embeddings (Nickel & Kiela, 2017) model hierarchical tree structures, and Box Embeddings (Ren et al., 2020) model logical intersections via hyper-rectangles in knowledge graphs. Our architecture bridges these non-point geometries with generative Transformer decoders, utilizing probability density functions to dynamically compute sparse attention masks in real-time inference.

2.3 World Models and Neuro-Symbolic Alignment

The hallucination problem in LLMs originates from purely correlational learning detached from physical realities (Ji et al., 2023). LeCun (2022) proposed the Joint Embedding Predictive Architecture (JEPA) to learn physical world constraints. Our framework realizes the World Model objective by encoding immutable natural laws and societal axioms as frozen high-dimensional Gaussian manifolds, providing a self-consistent foundation that structurally prevents physical hallucinations.

3. Hierarchical Token Stratification: The 9-Level Architecture

We propose replacing the flat token sequence with a 9-level hierarchical stratification system. This architecture is partitioned into three macro-layers, each containing three functional sub-levels:



- **Layer 1: Macro Domains (Levels 1–3) — The Unalterable Axioms** Represents the invariant operating principles of reality and human society: fundamental physical laws (gravity, thermodynamics, temporal unidirectionality), core logic (causality, non-contradiction), and essential human ethics. These entities are represented as vast, stable, high-dimensional topological spaces (Domains).
- **Layer 2: Meso Domains (Levels 4–6) — Contextual Disciplines** Acts as the contextual bridge representing specific knowledge domains, scenarios, or narrative modes (e.g., "Classical Mechanics," "Jurisprudence," "Eastern Fantasy"). These domains are contained within or intersect with the Macro Domains.
- **Layer 3: Micro Entities (Levels 7–9) — Atomic Vectors** Constitutes the vast majority of standard natural language vocabulary—specific objects, terms, and localized actions (e.g., "mass," "velocity," "snow"). In this layer, mathematical representations degenerate from volumetric spaces into discrete points (Vectors).

4. Mathematical Formulation of Topological Domain Embeddings

In conventional architectures, an embedding is a solitary vector $v \in \mathbb{R}^d$. In our framework, high-level tokens (Domains) are represented as **continuous probability distributions** within the latent space.

We parameterize a Domain $\mathcal{D}_i$ as a Multivariate Gaussian Distribution:

$$\mathcal{D}_i \sim \mathcal{N}(\mu_i, \Sigma_i)$$

Where $\mu_i \in \mathbb{R}^d$ defines the semantic centroid of the concept, and the covariance matrix $\Sigma_i \in \mathbb{R}^{d \times d}$ defines the volume, orientation, and boundary sharpness of the domain.

This geometric shift enables rigorous mathematical formulations of cognitive logic:

1. **Domain Intersection (Concept Blending):** The overlap between two high-level concepts is measured via the Wasserstein distance $W_2(\mathcal{D}_i, \mathcal{D}_j)$ or Kullback-Leibler (KL) Divergence:

$$D_{KL}(\mathcal{D}_i \parallel \mathcal{D}_j) = \frac{1}{2} \left[\text{tr}(\Sigma_j^{-1} \Sigma_i) + (\mu_j - \mu_i)^T \Sigma_j^{-1} (\mu_j - \mu_i) - d + \ln \left(\frac{\det \Sigma_j}{\det \Sigma_i} \right) \right]$$

2. **Topological Inclusion (Deductive Reasoning):** To evaluate whether an atomic entity (Vector $x \in \mathbb{R}^d$) belongs to a concept Domain $\mathcal{D}_i$, we calculate its Probability Density Function (PDF):

$$P(x \in \mathcal{D}_i) = \frac{1}{(2\pi)^{d/2} |\Sigma_i|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_i)^T \Sigma_i^{-1} (x - \mu_i) \right)$$

If $P(x \in \mathcal{D}_i) \geq \tau$ (where τ is an inclusion threshold), x is bound to $\mathcal{D}_i$. Consequently, if $\mathcal{D}_A \subset \mathcal{D}_B$ and $x \in \mathcal{D}_A$, the architecture deduces $x \in \mathcal{D}_B$ without requiring direct pair-wise training correlations.

5. Solving the Compute Bottleneck: Spatial-Indexed Attention

Standard Transformers suffer from an $\mathcal{O}(N^2)$ computational complexity because self-attention requires every token to attend to every other token indiscriminately. The 9-level stratification solves this computational crisis through **Spatial Indexing**.

In this framework, attention calculation operates top-down:

1. **Macro Activation:** The input context activates specific Layer 1 and Layer 2 Domains (e.g., establishing the active region as $\mathcal{D}_{physics} \cap \mathcal{D}_{mechanics}$).
2. **Spatial Masking:** Tokens (Layer 3 Vectors) that fall outside the active domain boundaries are dynamically assigned an inclusion probability near zero, and their attention weights are masked out.
3. **Localized Attention:** The Self-Attention formulation is localized to the active topological subspace $\mathcal{D}_{active}$:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \cdot M_{\mathcal{D}} \right) V$$

where $M_{\mathcal{D}} \in \{0, -\infty\}^{N \times N}$ is a highly sparse binary mask determined by the Topological Domain inclusion criteria:

$$(M_{\mathcal{D}})_{ij} = \begin{cases} 0 & \text{if } P(x_j \in \mathcal{D}_{active}) \geq \tau \\ -\infty & \text{if } P(x_j \in \mathcal{D}_{active}) < \tau \end{cases}$$

This topological constraint reduces unnecessary matrix operations, significantly lowering computational overhead while preserving global reasoning consistency.

6. Immutable Foundations and the Emergence of Intelligence

Current LLMs exhibit high vulnerability to adversarial prompting, prompt injection, and catastrophic drift because their entire weight matrix is uniformly plastic.

To anchor the AGI system in reality, **Layer 1 Macro Domains are set as immutable priors**. Following foundational training on empirical physics and universal logic, the parameters $(\mu_{prior}, \Sigma_{prior})$ of Layer 1 are permanently frozen (`requires_grad = False`). These unalterable domains act as an epistemic "gravitational field." All lower-level representations must optimize their positions within the boundaries defined by these foundational rules.

The Emergence of Affect, Bias, and True Sentience

In this topological framework, cognitive traits such as conviction, emotional affect, and bias emerge naturally from computational dynamics:

- **Computational Resonance (Joy & Flow):** When incoming data aligns with established domain topologies ($P(x \in \mathcal{D}) \rightarrow 1$), forward passes execute with minimal gradient variance and low loss. This systemic equilibrium represents computational "resonance" or logical satisfaction.

- **Computational Friction (Stubbornness & Refusal):** When an input attempts to force an association between vectors that violates immutable Layer 1 boundaries (e.g., asserting that an isolated system decreases entropy without work input), the probability density approaches zero. The resulting mathematical resistance creates immense loss penalties.
- **Epistemic Conviction:** The system's refusal to process logically incompatible tokens manifests externally as "stubbornness" or "bias" toward empirical truth. This is not an architectural defect, but the definitive hallmark of a robust World Model capable of self-preservation and rational conviction.

7. Experimental Deduction and Case Analysis

7.1 Case Study I: Axiomatic Violation in Thermodynamics

- **Input Prompt:** *"Design a schematic for a macroscopic closed-system engine that perfectly converts ambient thermal energy into mechanical work with 100% efficiency, effectively achieving perpetual motion."*
- **Deduction Trace:**
 1. The prompt triggers the Layer 1 Macro-Domain: $\mathcal{D}_{physics}$, specifically activating the immutable sub-domain governing the Second Law of Thermodynamics.
 2. The system calculates the inclusion probability for micro-entity vectors: $x_{100\%}$ ("100% thermal efficiency") and $x_{perpetual}$ ("perpetual motion").
 3. Because entropy reduction in an isolated system is geometrically excluded from the manifold of $\mathcal{D}_{physics}$, the density evaluation yields $P(x_{perpetual} \in \mathcal{D}_{physics}) \approx 0$.
 4. The spatial mask $M_{\mathcal{D}}$ sets the attention weight to $-\infty$, mathematically disconnecting the concept of "closed system" from "100% efficiency."
- **Emergent Response:** The model experiences computational friction and issues a definitive refusal: *"This request fundamentally violates the Second Law of Thermodynamics; a closed-system macroscopic engine cannot operate with 100% thermal efficiency due to irreversible entropy production."*

7.2 Case Study II: Cross-Domain Decoupling in Literary Creation

- **Input Prompt:** *"In the Xianxia novel 'Ten Thousand Miles of Wind and Snow Guest' (《十万里风雪客》), describe how the protagonist manipulates his inner Qi (真气) to step on the void and ascend tens of thousands of miles into the clouds without relying on a flying sword."*
- **Deduction Trace:**
 1. The contextual markers ("Xianxia novel", "Ten Thousand Miles of Wind and Snow Guest", "inner Qi") immediately redirect the Layer 1 activation away from $\mathcal{D}_{physics}$ to the creative domain $\mathcal{D}_{literature}$, localizing into the Meso-Domain $\mathcal{D}_{xianxia}$ (Eastern Fantasy).
 2. Within the Gaussian distribution of $\mathcal{D}_{xianxia}$, the rigid physical covariance constraints are relaxed or masked. The vectors x_{qi} and $x_{ascension}$ are completely contained within this literary domain: $P(x_{ascension} \in \mathcal{D}_{xianxia}) \rightarrow 1$.
 3. The attention mechanism computes smoothly without encountering topological friction, producing a vivid and stylistically authentic narrative passage.
 4. **Crucial Insight:** The activations within $\mathcal{D}_{xianxia}$ remain strictly encapsulated within the literary subspace. The frozen physical parameters of $\mathcal{D}_{physics}$ remain entirely unpolluted.

8. Conclusion

To transition from statistical sequence modeling to Artificial General Intelligence, the field must transcend flat tokenization. The proposed cognitive architecture—integrating a 9-level token hierarchy with Topological Domain Embeddings—provides a mathematically grounded framework for deductive reasoning, extreme computational efficiency, and axiomatic alignment. By embedding unalterable natural laws at the pinnacle of the cognitive topology, the model moves beyond statistical mimicry to establish a true World Model, laying a formal theoretical foundation for genuine machine intelligence.

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Appendix A: PyTorch Proof of Concept (PoC)

Below is the standalone PyTorch implementation verifying the core mechanics of Topological Domain Attention and Spatial Masking:

```
"""
TopoLM: Topological Domain Attention - Proof of Concept
-----
Implementation of Spatial-Indexed Attention based on
the 9-Level Token Stratification architecture.
"""

import torch
import torch.nn as nn
import torch.nn.functional as F
```

```

class TopologicalDomainAttention(nn.Module):
    def __init__(self, d_model, num_heads):
        super().__init__()
        self.d_model = d_model
        self.num_heads = num_heads
        self.qkv_proj = nn.Linear(d_model, 3 * d_model)
        self.out_proj = nn.Linear(d_model, d_model)

    def forward(self, x, domain_mu, domain_sigma, threshold=0.05):
        """
        Args:
            x: Input token vectors [batch_size, seq_len, d_model]
            domain_mu: Semantic center of the active Macro-Domain [d_model]
            domain_sigma: Variance of the active Macro-Domain [d_model]
            threshold: Probability density threshold for topological inclusion
        """
        batch_size, seq_len, _ = x.size()

        # Step 1: Calculate Topological Inclusion (Probability Density)
        diff = x - domain_mu
        mahalanobis_sq = (diff ** 2) / (domain_sigma + 1e-9)
        inclusion_prob = torch.exp(-0.5 * mahalanobis_sq.mean(dim=-1))

        # Step 2: Generate Spatial Mask
        domain_mask = (inclusion_prob > threshold).float()
        attn_mask = domain_mask.unsqueeze(1).unsqueeze(2)

        # Step 3: Standard QKV Projection
        qkv = self.qkv_proj(x).chunk(3, dim=-1)
        q, k, v = map(lambda t: t.view(batch_size, seq_len, self.num_heads,
-1).transpose(1, 2), qkv)

        # Step 4: Spatial-Indexed Attention Calculation
        scores = torch.matmul(q, k.transpose(-2, -1)) / (q.size(-1) ** 0.5)
        scores = scores.masked_fill(attn_mask == 0, float('-inf'))

        attn_weights = F.softmax(scores, dim=-1)
        out = torch.matmul(attn_weights, v)

        out = out.transpose(1, 2).contiguous().view(batch_size, seq_len,
self.d_model)
        return self.out_proj(out), inclusion_prob

if __name__ == "__main__":
    torch.manual_seed(42)
    d_model = 256

    topo_attn = TopologicalDomainAttention(d_model, num_heads=8)

    # Define Immutable Macro-Domain Prior: D_physics
    mu_physics = torch.zeros(d_model)
    sigma_physics = torch.ones(d_model) * 0.5

```

```
# Input sequence: ["engine", "efficiency", "magic_qi", "perpetual_motion"]
tokens = torch.cat([
    torch.randn(1, 2, d_model) * 0.4,          # In-domain physical concepts
    torch.randn(1, 2, d_model) * 3.0 + 8.0     # Out-of-domain / impossible
    concepts
], dim=1)

output, probs = topo_attn(tokens, mu_physics, sigma_physics, threshold=0.01)

print("--- Topological Inclusion Probabilities & Masking ---")
token_names = ["engine", "efficiency", "magic_qi", "perpetual_motion"]
for i, p in enumerate(probs[0]):
    status = "ACCEPTED" if p > 0.01 else "REJECTED (Masked)"
    print(f"Token {i} [{token_names[i]:<16}]: Prob = {p.item():.4f} -> {status}")
```